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Report

On

Green, Energy & Environment Audit For

Academic Year 2022 – 2023

of

**Bharatiya Vidya Bhavan's
Sardar Patel Institute of
Technology** Mumbai 400 058

Prepared

By

**Senergy Consultants Pvt
Ltd** Mumbai 400 088

March – 2023

Helping You to Conserve Energy

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SENERGY CONSULTANTS PVT LTD

Ref: SCPL-PR-918-310323 Date: March 31, 2023

Green, Energy and Environment Audit Report

Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology

I

Introduction

Green, Energy & Environment Audit was undertaken at Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology, Bhavan's Campus, Munshi Nagar, Andheri West, Mumbai - 400 058 during the month of March 2023.

The organization is very keen to promote green culture wherever possible, as a commitment towards better environment and conservation of energy. A lot of efforts have already been put up to bring down the carbon footprint. To further optimize consumption and identify saving opportunities, M/s Senergy Consultants was assigned to carry out Green & Environment Audit of the premises.

This Audit Report presents the analysis of the data collected, observations made at the facility and is governed by the objectives, scope of work, methodology etc. discussed in the ensuing paragraphs.

Team:

The team members of the audit study.

- Mr. Ravindra Datar
- Mr. Nitesh Kharche
- Mr. Tushar Kamble

Acknowledgment:

We wish to express our gratitude towards Dr. B. N. Chaudhari for having given us the opportunity for conducting the study and the support provided during the study.

We are also thankful to Prof. Ananthalakshmi for extending necessary help and co-operation from their side.

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II

Executive Summary

The premises were evaluated against the various criteria laid down by the National Assessment and Accreditation Council (NAAC). The major observations are.

Air Quality & Ventilation:

- The classrooms and other area are well ventilated to ensure proper air quality.
 - The fans are appropriately installed to ensure proper air circulation
 - The indoor as well as outdoor plants have also been provided to improve the environment. •
- The air-conditioned rooms are provided with proper ventilation and fresh air.

Lighting System:

- The usage of natural light is optimized through well designed structure and windows.
 - It is suggested to automate switching of lamps in the common areas and washrooms with sensor based control. •
- Most of the light fitting are provided with high efficiency LED lamps. The remaining lamps are being replaced with LED lamps.

Green Campus Initiative:

- The movement of vehicle inside the campus is restricted with vehicles of Staff, Special Dignitaries and VIP Guests are allowed to enter the campus.
 - Pedestrian friendly pathways have been constructed for easy movement inside the campus. •
- There is a complete ban plastic usage inside the campus.
- The campus is surrounded with a lot of greenery, trees, and proper landscaping.

Environment & Energy Initiative:

- The college has formed internal committees and student groups / clubs for promoting environmental and conservation activities.
- Green Consumerism and Green Entrepreneurship workshops are conducted to create awareness related to the environment.
- Various Tree Plantation programs have been carried out within and outside the campus.

Water Quality & Conservation:

- The water is supplied by the Municipal Corporation, which is a common practice in Mumbai, Thane & Navi Mumbai.
- Water purifiers & coolers are provided at convenient locations.
- The distribution network and piping are more or less satisfactory and adequate.
- The toilets are provided with water efficient (low usage) fittings.
- The rainwater harvesting pit is provided for ground water recharge.

Waste Management:

- The effluent water is discharged in the common municipal drain, which is a common practice in and around Mumbai Thane & Navi Mumbai.

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- The organic waste is segregated and converted into manure in the composting pits.
- The electronic gadgets / waste is either donated if useful or handed over to appropriate waste collectors.
- The general solid waste is disposed through municipal corporation.

Green IT culture:

- The Energy efficient computers and laptops have been procured.
- The electronic communication is encouraged to minimize usage of papers.
- Most of the papers are reused for doubled sided printing to further minimize usage of paper.

Air Conditioning System:

- The Air Conditioners are operated as required with manual control. The operation is minimal consequently automation may not be economical.
- The room temperature is maintained at 24 to 25 °C, which is well within the recommended values.
- The Air Conditioners are serviced regularly and properly maintained.
- Most of the Air Conditioners units are energy efficient with 3 star rating.
- The air-conditioned space is properly sealed to prevent air ingress / cold air loss.

Infrastructure usage:

- Ramps are provided on the ground floor to address the needs of differently abled persons.
- The on-campus movement is distributed with multiple entrances as well as staircases.
- The fire extinguishers are provided at all key areas.
- The college has initiated appropriate measures to meet the safety requirement.
- The draining system for washrooms is efficient and effective.
- There were no seepages observed in the building premises.

Renewable Energy:

- The college has installed 3kWp Solar Photovoltaic System to operate the water pump. Also, they are installing Solar Photovoltaic System (11kWp x 3nos.) with NET metering to meet part of the electricity consumption.
- The quantity of plate waste (organic waste with higher starch contents) is negligible, consequently, there is no potential for biogas generation.

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Green, Energy and Environment Audit Report

Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology

III

Electrical System

Gadget List

Lamps:

Location	LED
	22 W
Ground Floor	
Room No 001	12
Room No 002	20
Room No 003	25
Room No 004,005	45
Room No 006	20
Room No 008	24
Room No 009	12
Ground Floor Passage & Bathroom	40
First Floor	
Room No 101	12
Room No 102	16
Room No 103	12
Room No 105	24
Room No 106	12
Room No 107	16
Room No 108	17
1 st Floor & Toilet	20
Second Floor	
Room No 201	12
Room No 202	16

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Room No 203	12
Room No 204	60
Room No 205	12
Room No 206	16
Room No 207	12
2 nd Floor & Toilet	16
Third Room	
Room No 301	12
Room No 302	16
Room No 303	12
Room No 304	12
Room No 305	8
Room No 306	6
Room No 307	6
Room No 308	6
Room No 309	7
Room No 310	15
Room No 311	15
Room No 312	12
3 rd Floor & Toilet	28

Location	LED
	22 W
Fourth Floor	
Room No 401	7
Room No 402	15
Room No 403	32
Room No 405	8

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Room No 406	12
Room No 407	6
Room No 408	8
Room No 409	6
Room No 410	27
Room No 411	14
Room No 412	10
4 th Floor & Toilet	22
Fifth Floor	
Room No 501	8
Room No 502	15
Room No 503	20
Room No 504	6
Room No 505	6
Room No 506	9
Room No 507 A/B	8
Room No 508	6
Room No 509	6
5 th Floor & Toilet	25
Sixth Floor	
Room No 601	8
Room No 602	24
Room No 603	32
Room No 604	5
Room No 605	8
Room No 606	24
Room No 607 A & B	10
Room No 608	24

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Room No 609	6
6 th Floor & Toilet	25
Seventh Floor	
Room No 701	8
Room No 702	26
Room No 703	12
7 th Floor & Toilet	10
Eighth Floor	
Room No 8th Floor	60
8 th Floor & Toilet	10

Ceiling Fans:

Location	Quantity (No.)
Ground Floor	
Room No 001	9
Room No 002	9
Room No 004,005	2
Room No 006	11
Room No 009	9
Ground Floor Passage & Bathroom	4
First Floor	
Room No 101	9
Room No 102	13
Room No 103	9
Room No 106	9
Room No 107	11
Room No 108	9
Second Floor	

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Room No 201	9
Room No 202	12
Room No 203	9
Room No 204	31
Room No 205	6
Room No 206	12
Room No 207	9
Third Floor	
Room No 301	9
Room No 302	12
Room No 303	6
Room No 304	4
Room No 305	6
Room No 306	4
Room No 307	7
Room No 308	4
Room No 309	7
Room No 310	9
Room No 311	12
Room No 312	9
3 rd Floor	2
Fourth Floor	
Room No 401	7
Room No 402	12
Room No 403	9
Room No 405	7
Room No 407	7
Room No 409	7

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Room No 410	8
Room No 411	12
Room No 412	6
4 th Floor	3
Fifth Floor	
Room No 501	8

Location	Quantity (No.)
Room No 502	12
Room No 503	8
Room No 504	4
Room No 505	9
Room No 506	6
Room No 507 A/B	6
Room No 508	4
Room No 509	9
Sixth Floor	
Room No 601	8
Room No 602	12
Room No 603	9
Room No 604	9
Room No 605	9
Room No 606	6
Room No 607 A & B	10
Room No 608	5
Room No 609	7
Seventh Floor	
Room No 701	9

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Room No 702	8
Room No 703	9
7 th Floor & Toilet	EX 2
Eighth Floor	
Room No 8th Floor	8
8 th Floor	2

Air Conditioning Units:

Location	Type	Capacity (TR)	Quantity (Nos.)
Ground Floor			
Room No 003	Split	5.5	3
Room No 004	Split	8.5	1
Room No 005	Split	3	1
Room No 008	Split	2	7
Room No 009	Split	2	2
First Floor			
Room No 101	Split	3	2
Room No 102	Split	2	5
Room No 104	Split	2	4
Room No 105	Split	2	3
Room No 108	Split	3	2
Second Floor			
Room No 201	Split	3	2
Room No 203	Split	2	4
Room No 207	Split	3	2
Third Room			
Room No 303	Split	2	3
Room No 305	Split	3	2

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Room No 307	Split	3	2
Room No 308	Split	2	2

Location	Type	Capacity (TR)	Quantity (Nos.)
Room No 309	Split	3	2
Room No 310	Split	2	3
Fourth Floor			
Room No 401	Split	3	1
Room No 403	Split	2	4
Room No 404	Split	2	2
Room No 405	Split	3	2
Room No 406	Split	2	3
Room No 407	Split	3	2
Room No 408	Split	2	2
Room No 409	Split	3	2
Room No 410	Split	2	3
Room No 412	Split	2	3
Fifth Floor			
Room No 501	Split	3	2
Room No 503	Split	2	2
Room No 505	Split	3	2
Room No 506	Split	2	1
Room No 507 A/B	Split	2	1
Room No 508	Split	2	2
Room No 509	Split	3	2
Sixth Floor			
Room No 601	Split	3	2
Room No 602	Split	2	5

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Room No 604	Split	2	4
Room No 605	Split	3	2
Room No 609	Split	3	2
Seventh Floor			
Room No 701	Split	3	2
Room No 702	Split	2	4
Eighth Floor			
Apple Room	Split	3	2
Staff Room	Split	3	1
Conference Room	Split	3	1
Startup Room	Split	2	1
Room 1	Split	2	1
Room2	Split	2	1
Room 3	Split	2	1
Room 4	Split	2	1
Room 5	Split	2	1
Room 6	Split	2	1
Reception Passage	Split	1	2

Observations & Suggestions:

- Most of the light fitting are provided with high efficiency LED lamps. The remaining lamps are being replaced with LED lamps.
- The fans are of standard efficiency; which may be replaced with high efficiency fans with BLDC motor.
- The Air Conditioning Units are with 3-star rating, which are intrinsically operate at high energy efficiency.
- The Air Conditioners are operated as required with manual control. The operation is minimal consequently automation may not be economical.
- The temperature is maintained at 24 to 25 °C for air conditioning spaces, which is within the recommended values.
- The rooms are well ventilated and provided with fans at appropriate location for proper air circulation.
- The gadgets are services properly and maintained in good condition.

Electricity Bills:

- The power cost is within the normal range.

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IV

Environmental System

Ventilation & Air Quality:

- The air ventilation is adequate.
- Several indoor & outdoor plants have been installed to improve air quality.
- The air-conditioned rooms are provided with proper ventilation and fresh air.
- It has been a general practice to switch off the fans & lights in an unoccupied area.



Latitude - 19.123
Longitude - 72.83
4RFP+9C9 Bhavan's Campus, Old D N
Nagar, Munshi Nagar, Andheri West,
Mumbai, Maharashtra 400058, India.
31/03/23 14:40 PM GMT +05:30

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V

Water Management

Water Source:

The water supplied by the municipal corporation is used for drinking and other requirements. The incoming water from the municipal corporation is metered.

Observation & Suggestions:

- The cost as well as consumption is within the normal range.

Water Purifiers:

Sr No	Location	Quantity (No.)
1	Ground Floor	2
2	First Floor	1
3	Second Floor	1
4	Third Floor	1
5	Fourth Floor	1
6	Fifth Floor	1
7	Sixth Floor	1

The water purifiers are provided at convenient locations.

Water Distribution System:

The distribution network and piping are satisfactory and adequate.

Rainwater Harvesting:

The rainwater is fed to the rainwater harvesting pit for ground water recharge.



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VI

Waste Generation & Management

Solid Waste:

- The organic as well as inorganic waste is segregated in the college premises.
- The organic waste is used to generate manure by composting.





- There is no generation of hazardous or biomedical waste in the premises.
- Each Floor is provided is with segregated dustbin for Recyclable waste and Non-Recyclable Waste
- The electronic gadgets with residual life are donated while the electronic waste is properly segregated and handed over to appropriate scrap collector.

Sewage & Wastewater:

- The sewage is disposed through municipal system.
- There is no generation of harmful or hazardous effluent.
- The sewage treatment plant is not required due to minimal generation of waste.
- The municipal corporation charges cess for treating and disposing wastewater.

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VII

Infrastructure & Safety



Latitude - 19.123
Longitude - 72.83
4RFP+9C9 Bhavan's Campus, Old D N
Nagar, Munshi Nagar, Andheri West,
Mumbai, Maharashtra 400058, India.
31/03/23 14:20 PM GMT +05:30

Observations:

- The premises are provided with multiple staircases with necessary entrances to ensure quick and effective movement in normal as well as emergency conditions.
- The movement of vehicle inside the campus is restricted with vehicles of Special Dignitaries and VIP Guests are allowed to enter the campus.
- There are pedestrian friendly pathways for in-campus movement.
- There is restriction on the usage of plastic, which may be extended to completely ban plastic usage inside the campus.
- There is adequate parking space for vehicles/four wheelers. However, almost all the students and many of the faculty members avail public transport system which is very convenient due to proximity to railway station and bus services.
- Ramp & wheelchair is available for differently abled student.

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Draining system:

- The drains from the washrooms are connected to the municipal drainage, which is a common practice in the colleges in and around Mumbai.

Seepage in the building:

- The premise was visually inspected for seepages. No seepages were observed in any of the places.

Firefighting & fire escape system:

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Sr No	Location	Type	Capacity	Quantity (No.)
1	Ground Floor	ABC Dry Powder	6 KG	5
2	First Floor	ABC Dry Powder	6 KG	2
		ABC Safe Zone	5 KG	3
3	Second Floor	ABC Dry Powder	6 KG	7
		ABC Safe Zone	5 KG	1
		Water	9 LTR	2
4	Third Floor	ABC Dry Powder	6 KG	5
		Water	9 LTR	1
5	Fourth Floor	ABC Dry Powder	6 KG	5
		ABC Safe Zone	5 KG	1
6	Fifth Floor	ABC Dry Powder	6 KG	1
		ABC Safe Zone	5 KG	3
		Water	9 LTR	1
7	Sixth Floor	ABC Dry Powder	6 KG	4
		Water	9 LTR	1
8	Seventh Floor	ABC Dry Powder	6 KG	1
9	Eighth Floor	ABC Dry Powder	6 KG	2

- There are efficient fire extinguishers in the premises, which are checked / refilled as per the stipulated frequency.
- The premise is provided with multiple staircases with requisite entrances to ensure quick and effective movement in emergency conditions.

Parking space:



- There is adequate parking space for vehicles/four wheelers. However, almost all the students and many of the faculty members avail public transport system which is very convenient due to proximity to railway station and bus services.

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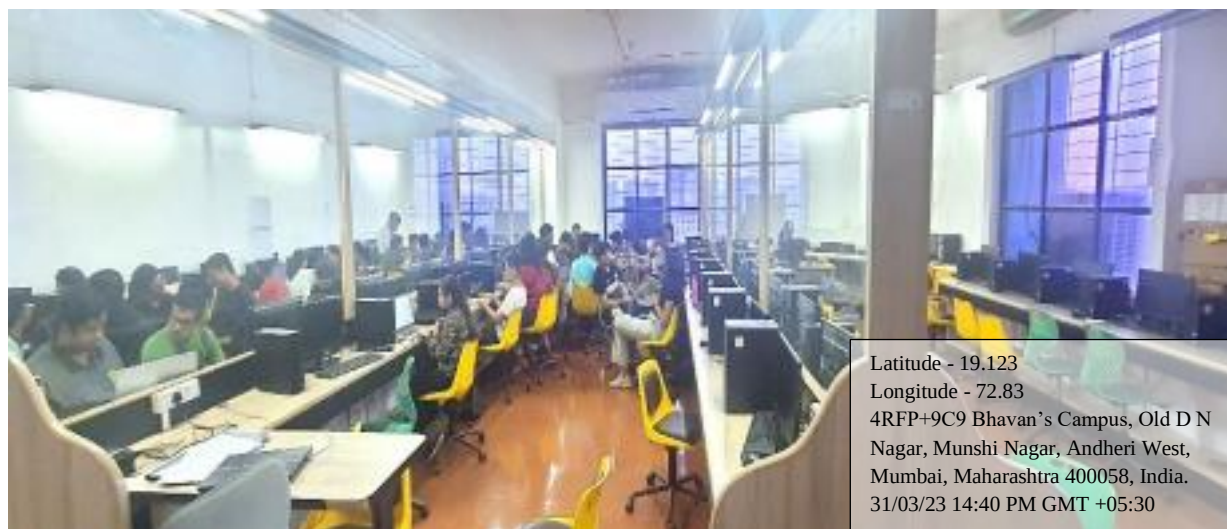
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VIII Green Culture



Observations & Suggestions:

- The LED / LCD monitors & Laptops has been procured, which are energy efficient.
- These monitors are not only energy efficient but also generate minimal heat and cut down on air conditioning load.
- The electronic communication is encouraged to minimize usage of papers.
- Most of the papers are reused for doubled sided printing to further minimize usage of paper.

The following steps may be initiated to further enhance efficiency of the systems.

1. An efficient power management system may be incorporated to
 - a. Switch off the display if not in use.
 - b. Put the computer in Sleep mode / switching off the machines, if not used for prolonged period.
2. Optimize brightness of the screen.
3. Discourage use of screen savers, which has similar power consumption.

Paper-less communication:

The major internal as well as external communication is through electronic medium.

Re-using one sided paper for printing:

It was observed that two side printing / printing on the back side of used paper in more than 80% of the cases.

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IX Renewable Energy

Solar Photovoltaic:



The college has installed 3kWp Solar Photovoltaic System to operate the water pump. Also, they are installing Solar Photovoltaic System (11kWp x 3nos.) with NET metering to meet part of the electricity consumption.

Solar Thermal:

There is no application of solar thermal system and does not find attractive in this case.

Biogas Plant:

There is no possibility of installing biogas plant for cooking as the quantity of plate waste is negligible.

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I

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- Mr. Tushar Kamble

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The premises were evaluated against the various criteria laid down by the National Assessment and Accreditation Council (NAAC). The major observations are.

Renewable Energy:

- The possibility of installing Solar Photovoltaic System with NET metering may be assessed to meet part of the electricity consumption.
- The quantity of plate waste (organic waste with higher starch contents) is negligible, consequently, there is no potential for biogas generation.

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- The classrooms and other area are well ventilated to ensure proper air quality.
- The fans are appropriately installed to ensure proper air circulation
- The indoor as well as outdoor plants have also been provided to improve the environment.
- The air conditioned rooms are provided with proper ventilation and fresh air.

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- The usage of natural light is optimized through well designed structure and windows.
- It is suggested to automate switching of lamps in the common areas and washrooms with sensor based control.
- Most of the light fittings are provided with high efficiency LED lamps. The conventional tube lights operating for longer period should be replaced with energy efficient LED lamps.

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- The movement of vehicle inside the campus is restricted with vehicles of Staff, Special Dignitaries and VIP Guests are allowed to enter the campus.
- Pedestrian friendly pathways have been constructed for easy movement inside the campus.
- There is complete ban plastic usage inside the campus.
- The campus is surrounded with a lot of greenery, trees, and proper landscaping.

Environment & Energy Initiative:

- The college has formed internal committees and student groups / clubs for promoting environmental and conservation activities.
- Various Tree Plantation programs have been carried out within and outside the campus.

Water Quality & Conservation:

- The water is supplied by the Municipal Corporation, which is a common practice in Mumbai, Thane & Navi Mumbai.
- Water purifiers & coolers are provided at convenient locations.
- The distribution network and piping are more or less satisfactory and adequate.
- The toilets are provided with water efficient (low usage) fittings.
- The rainwater harvesting pit is provided for ground water recharge.

Waste Management:

- The effluent water is discharged in the common municipal drain, which is a common practice in and around Mumbai Thane & Navi Mumbai.

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- The organic waste is segregated and converted into manure in the composting pits.
- The general solid waste is disposed through municipal corporation.

Green IT culture:

- The Energy efficient computers and laptops have been procured.
- Most of the papers are reused for doubled sided printing to further minimize usage of paper.
- The electronic communication may be encouraged to minimize usage of papers.

Air Conditioning System:

- The Air Conditioners are operated as required with manual control. The operation is minimal consequently automation may not be economical.
- The room temperature is maintained at 24 to 25 °C, which is well within the recommended values.
- The Air Conditioners are serviced regularly and properly maintained.
- Most of the Air Conditioners units are energy efficient with 3 star rating.
- The air conditioned space is properly sealed to prevent air ingress / cold air loss.

Infrastructure usage:

- Ramps are provided on the ground floor to address the needs of differently abled persons.
- The on-campus movement is distributed with multiple entrances as well as staircases.
- The fire extinguishers are provided at all key areas.
- The college has initiated appropriate measures to meet the safety requirement.
- The draining system for washrooms is efficient and effective.
- There were no seepages observed in the building premises.

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III Electrical System

Gadget List

Lamps:

Location	LED	TFL
	22 W	28 W
Ground Floor		
Room No 001	7	5
Room No 002	11	9
Room No 003	14	11
Room No 004,005	25	20
Room No 006	11	9
Room No 008	14	10
Room No 009	7	5
Ground Floor Passage & Bathroom	22	18
First Floor		
Room No 101	7	5
Room No 102	9	7
Room No 103	7	5
Room No 105	14	10
Room No 106	7	5
Room No 107	9	7
Room No 108	10	7
1 st Floor & Toilet	11	9
Second Floor		
Room No 201	7	5
Room No 202	9	7
Room No 203	7	5
Room No 204	33	27
Room No 205	7	5
Room No 206	9	7
Room No 207	7	5
2 nd Floor & Toilet	9	7
Third Room		
Room No 301	7	5
Room No 302	9	7
Room No 303	7	5
Room No 304	7	5
Room No 305	4	4
Room No 306	3	3
Room No 307	3	3
Room No 308	3	3
Room No 309	4	3
Room No 310	9	6
Room No 311	9	6
Room No 312	7	5
3 rd Floor & Toilet	16	12

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Location	LED	TFL
	22 W	28 W
Fourth Floor		
Room No 401	4	3
Room No 402	9	6
Room No 403	18	14
Room No 405	5	3
Room No 406	7	5
Room No 407	3	3
Room No 408	5	3
Room No 409	4	2
Room No 410	15	12
Room No 411	8	6
Room No 412	6	4
4 th Floor & Toilet	12	10
Fifth Floor		
Room No 501	5	3
Room No 502	9	6
Room No 503	11	9
Room No 504	3	3
Room No 505	3	3
Room No 506	5	4
Room No 507 A/B	5	3
Room No 508	3	3
Room No 509	3	3
5 th Floor & Toilet	14	11
Sixth Floor		
Room No 601	5	3
Room No 602	14	10
Room No 603	18	14
Room No 604	3	2
Room No 605	5	3
Room No 606	14	10
Room No 607 A & B	6	4
Room No 608	14	10
Room No 609	3	3
6 th Floor & Toilet	14	11
Seventh Floor		
Room No 701	5	3
Room No 702	15	11
Room No 703	7	5
7 th Floor & Toilet	6	4
Eighth Floor		
Room No 8th Floor	33	27
8 th Floor & Toilet	6	4

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Ceiling Fans:

Location	Quantity (No.)
Ground Floor	
Room No 001	9
Room No 002	9
Room No 004,005	2
Room No 006	11
Room No 009	9
Ground Floor Passage & Bathroom	4
First Floor	
Room No 101	9
Room No 102	13
Room No 103	9
Room No 106	9
Room No 107	11
Room No 108	9
Second Floor	
Room No 201	9
Room No 202	12
Room No 203	9
Room No 204	31
Room No 205	6
Room No 206	12
Room No 207	9
Third Floor	
Room No 301	9
Room No 302	12
Room No 303	6
Room No 304	4
Room No 305	6
Room No 306	4
Room No 307	7
Room No 308	4
Room No 309	7
Room No 310	9
Room No 311	12
Room No 312	9
3 rd Floor	2
Fourth Floor	
Room No 401	7
Room No 402	12
Room No 403	9
Room No 405	7
Room No 407	7
Room No 409	7
Room No 410	8
Room No 411	12
Room No 412	6
4 th Floor	3
Fifth Floor	
Room No 501	8

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Location	Quantity (No.)
Room No 502	12
Room No 503	8
Room No 504	4
Room No 505	9
Room No 506	6
Room No 507 A/B	6
Room No 508	4
Room No 509	9
Sixth Floor	
Room No 601	8
Room No 602	12
Room No 603	9
Room No 604	9
Room No 605	9
Room No 606	6
Room No 607 A & B	10
Room No 608	5
Room No 609	7
Seventh Floor	
Room No 701	9
Room No 702	8
Room No 703	9
7 th Floor & Toilet	EX 2
Eighth Floor	
Room No 8th Floor	8
8 th Floor	2

Air Conditioning Units:

Location	Type	Capacity (TR)	Quantity (Nos.)
Apple Room	Split	3	2
Staff Room	Split	3	1
Conference Room	Split	3	1
Startup Room	Split	2	1
Reception Passage	Split	1	2

Observations & Suggestions:

- Most of the light fittings are provided with high efficiency LED lamps. The conventional tube lights operating for longer period should be replaced with energy efficient LED lamps.
- The Air Conditioning Units are with 3 star rating, which are intrinsically operate at high energy efficiency.
- The Air Conditioners are operated as required with manual control. The operation is minimal consequently automation may not be economical.
- The temperature is maintained at 24 to 25 °C for air conditioning spaces, which is within the recommended values.
- The fans are of standard efficiency; which may be replaced with high efficiency fans with BLDC motor.
- The rooms are well ventilated and provided with fans at appropriate location for proper air circulation.

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- The gadgets are services properly and maintained in good condition.

Electricity Bills:

Consumer Name: Sardar Patel Institute of Technology					Service Provider: Adani Electricity		
Tariff Category: LT IX (B)					Consumer No. 102694354		
Meter No.: L985263					Contract Demand: 180 KVA		
Description	Unit	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19
Energy Consumption	KWH	10631	11835	10772	10265	12545	9039
Bill	Rs	118970	132330	121170	96770	119480	102470
Cost	Rs/KWH	11.19	11.18	11.25	9.43	9.52	11.34
Description	Unit	Oct-19	Nov-19	Dec-19	Jan-20	Feb-20	Average
Energy Consumption	KWH	12306	11543	6986	9122	10578	10511
Bill	Rs	139390	130690	78910	102940	120710	114894
Cost	Rs/KWH	11.33	11.32	11.30	11.28	11.41	10.93

Consumer Name: Sardar Patel Institute of Technology					Service Provider: Adani Electricity		
Tariff Category: LT IX (B)					Consumer No. 152372652		
Meter No.: L1001405					Contract Demand: 129.60 KVA		
Description	Unit	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19
Energy Consumption	KWH	15056	8300	6463	9067	10523	7988
Bill	Rs	170320	93280	74020	102710	119090	90500
Cost	Rs/KWH	11.31	11.24	11.45	11.33	11.32	11.33
Description	Unit	Oct-19	Nov-19	Dec-19	Jan-20	Feb-20	Average
Energy Consumption	KWH	10846	10346	5395	6317	8070	8943
Bill	Rs	122890	117110	60650	71320	90880	101161
Cost	Rs/KWH	11.33	11.32	11.24	11.29	11.26	11.31

Consumer Name: Sardar Patel Institute of Technology					Service Provider: Adani Electricity		
Tariff Category: LT XI (B)					Consumer No. 152372205		
Meter No.: L1001160					Contract Demand: 127.80 KVA		
Description	Unit	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19
Energy Consumption	KWH	13319	7509	6563	9830	9574	7568
Bill	Rs	150070	84070	73850	111000	108200	85420
Cost	Rs/KWH	11.27	11.20	11.25	11.29	11.30	11.29
Description	Unit	Oct-19	Nov-19	Dec-19	Jan-20	Average	
Energy Consumption	KWH	10612	9281	4275	6750	8528	
Bill	Rs	119780	104640	48150	76810	96199	
Cost	Rs/KWH	11.29	11.27	11.26	11.38	11.28	

Observations & Suggestions:

- The power consumption & cost is within the normal range.

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IV Environmental System

Ventilation & Air Quality:

- The air ventilation is adequate.
- Several indoor & outdoor plants have been installed to improve air quality.



- The air conditioned rooms are provided with proper ventilation and fresh air.

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- It has been a general practice to switch off the fans & lights in an unoccupied area.



V

Water Management

Water Source:

The water supplied by the municipal corporation is used for drinking and other requirements. The incoming water from the municipal corporation is metered.



The water consumption details are as under.

Period	Days	Consumption	Bill Amount	Cost
		KL	Rs	Rs/KL
Connection No- KW@0359909				
20-04-2019 to 17-05-2019	27	456	3946	8.65
17-05-2019 to 17-06-2019	31	524	4542	8.67
17-07-2019 to 16-08-2019	30	507	4500	8.88
17-06-2019 to 17-07-2019	30	507	4500	8.88
16-08-2019 to 17-09-2019	32	541	4801	8.87
17-09-2019 to 17-10-2019	30	507	4500	8.88
17-10-2019 to 18-11-2019	32	541	4801	8.87
18-11-2019 to 16-12-2019	28	473	4197	8.87

Observation & Suggestions:

- The cost as well as consumption is within the normal range.

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Water Coolers & Purifiers:



Sr No	Location	Quantity (No.)
1	Ground Floor	2
2	First Floor	1
3	Second Floor	1
4	Third Floor	1
5	Fourth Floor	1
6	Fifth Floor	1
7	Sixth Floor	1

The water coolers & purifiers are provided at convenient locations.

Water Distribution System:

The distribution network and piping are more or less satisfactory and adequate.

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Rainwater Harvesting:

The rainwater is fed to the rainwater harvesting pit for ground water recharge.



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VI Waste Generation & Management

Solid Waste:

- The organic as well as inorganic waste is segregated in the college premises.
- The organic waste is used to generate manure by composting.



- There is no generation of hazardous or biomedical waste in the premises.

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- Each Floor is provided is with segregated dustbin for Recyclable waste and Non-Recyclable Waste



Sewage & Wastewater:

- The sewage is disposed through municipal system.
- There is no generation of harmful or hazardous effluent.
- The sewage treatment plant is not required due to minimal generation of waste.
- The municipal corporation charges cess for treating and disposing wastewater.

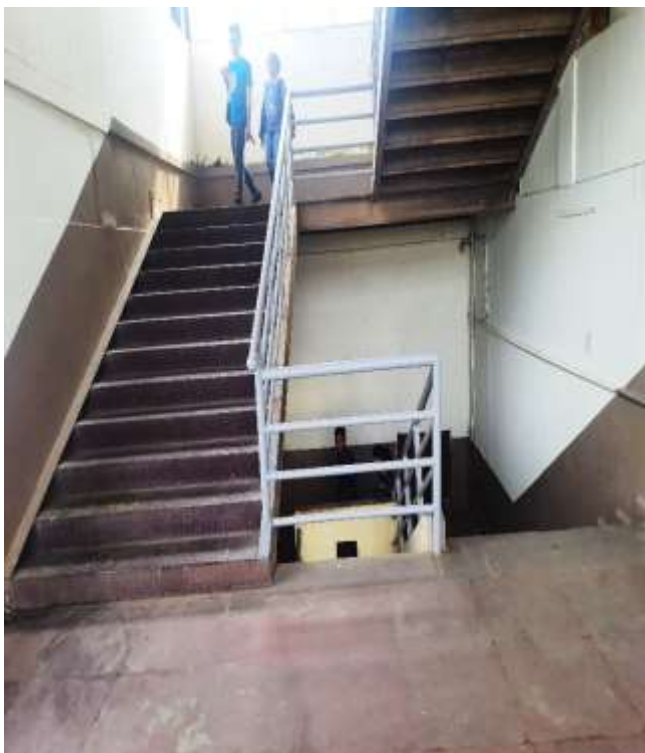
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VII Infrastructure & Safety



Observations:

- The premises are provided with multiple staircases with necessary entrances to ensure quick and effective movement in normal as well as emergency conditions.



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**Green, Energy and Environment Audit Report
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- The movement of vehicle inside the campus is restricted with vehicles of Special Dignitaries and VIP Guests are allowed to enter the campus.
- There are pedestrian friendly pathways for in-campus movement.
- There is complete ban plastic usage inside the campus.
- There is adequate parking space for vehicles/four wheelers. However, almost all the students and many of the faculty members avail public transport system which is very convenient due to proximity to railway station and bus services.
- Ramp & wheelchair is available for differently abled student.



Draining system:

- The drains from the washrooms are connected to the municipal drainage, which is a common practice in the colleges in and around Mumbai.

Seepage in the building:

- The premise was visually inspected for seepages. No seepages were observed in any of the places.

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Firefighting & fire escape system:



Sr No	Location	Type	Capacity	Quantity (No.)
1	Ground Floor	ABC Dry Powder	6 KG	5
2	First Floor	ABC Dry Powder	6 KG	2
		ABC Safe Zone	5 KG	3
3	Second Floor	ABC Dry Powder	6 KG	7
		ABC Safe Zone	5 KG	1
		Water	9 LTR	2
4	Third Floor	ABC Dry Powder	6 KG	5
		Water	9 LTR	1
5	Fourth Floor	ABC Dry Powder	6 KG	5
		ABC Safe Zone	5 KG	1
6	Fifth Floor	ABC Dry Powder	6 KG	1
		ABC Safe Zone	5 KG	3
		Water	9 LTR	1
7	Sixth Floor	ABC Dry Powder	6 KG	4
		Water	9 LTR	1
8	Seventh Floor	ABC Dry Powder	6 KG	1
9	Eighth Floor	ABC Dry Powder	6 KG	2

- There are efficient fire extinguishers in the premises, which are checked / refilled as per the stipulated frequency.
- The premise is provided with multiple staircases with requisite entrances to ensure quick and effective movement in emergency conditions.

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Parking space:



- There is adequate parking space for vehicles/four wheelers. However, almost all the students and many of the faculty members avail public transport system which is very convenient due to proximity to railway station and bus services.

VIII Green Culture



Observations & Suggestions:

- The LED / LCD monitors & Laptops has been procured, which are energy efficient.
- These monitors are not only energy efficient but also generate minimal heat and cut down on air conditioning load.
- The electronic communication may be encouraged to minimize usage of papers.
- Most of the papers are reused for doubled sided printing to further minimize usage of paper.

The following steps may be initiated to further enhance efficiency of the systems.

1. An efficient power management system may be incorporated to
 - a. Switch off the display if not in use.
 - b. Put the computer in Sleep mode / switching off the machines, if not used for prolonged period.
2. Optimize brightness of the screen.
3. Discourage use of screen savers, which has similar power consumption.

Re-using one sided paper for printing:

It was observed that two side printing / printing on the back side of used paper in more than 80% of the cases.

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IX Renewable Energy

Solar Photovoltaic:

It is suggested to install rooftop solar panels with net metering for captive usage.

Solar Thermal:

There is no application of solar thermal system and does not find attractive in this case.

Biogas Plant:

There is no possibility of installing biogas plant for cooking as the quantity of plate waste is negligible.

**Green, Energy and Environment Audit Report
Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology**

Report

On

Green, Energy & Environment Audit

For

Academic Year 2018 – 2019

of

**Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
Mumbai 400 058**

Prepared

By

**Senergy Consultants Pvt Ltd
Mumbai 400 088**

March – 2019

Helping You to Conserve Energy

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I Introduction

Green, Energy & Environment Audit was undertaken at Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology, Bhavan's Campus, Munshi Nagar, Andheri West, Mumbai - 400 058 during the month of March 2019.

The organization is very keen to promote green culture wherever possible, as a commitment towards better environment and conservation of energy. A lot of efforts have already been put up to bring down the carbon footprint. To further optimize consumption and identify saving opportunities, M/s Senergy Consultants was assigned to carry out Green & Environment Audit of the premises.

This Audit Report presents the analysis of the data collected, observations made at the facility and is governed by the objectives, scope of work, methodology etc. discussed in the ensuing paragraphs.

Team:

The team members of the audit study.

- Mr. Ravindra Datar
- Mr. Tushar Kamble

Acknowledgment:

We wish to express our gratitude towards Dr. Prachi Gharpure (Principal) for having given us the opportunity for conducting the study and the support provided during the study.

II

Executive Summary

The premises were evaluated against the various criteria laid down by the National Assessment and Accreditation Council (NAAC). The major observations are.

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- There is complete ban plastic usage inside the campus.
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- Various awareness programs have been carried out to promote 'Save Environment'.
- Various Tree Plantation programs have been carried out within and outside the campus.

Water Quality & Conservation:

- The water is supplied by the Municipal Corporation, which is a common practice in Mumbai, Thane & Navi Mumbai.
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III Electrical System

Gadget List

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	28 W	22 W
Ground Floor		
Room No 001	8	4
Room No 002	12	8
Room No 003	15	10
Room No 004,005	26	19
Room No 006	12	8
Room No 008	15	9
Room No 009	8	4
Ground Floor Passage & Bathroom	23	17
First Floor		
Room No 101	8	4
Room No 102	10	6
Room No 103	8	4
Room No 105	15	9
Room No 106	8	4
Room No 107	10	6
Room No 108	11	6
1 st Floor & Toilet	12	8
Second Floor		
Room No 201	8	4
Room No 202	10	6
Room No 203	8	4
Room No 204	34	26
Room No 205	8	4
Room No 206	10	6
Room No 207	8	4
2 nd Floor & Toilet	10	6
Third Room		
Room No 301	8	4
Room No 302	10	6
Room No 303	8	4
Room No 304	8	4
Room No 305	5	3
Room No 306	4	2
Room No 307	4	2
Room No 308	4	2
Room No 309	5	2
Room No 310	10	5
Room No 311	10	5
Room No 312	8	4
3 rd Floor & Toilet	17	11

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Location	TFL	LED
	28 W	22 W
Fourth Floor		
Room No 401	5	2
Room No 402	10	5
Room No 403	19	13
Room No 405	6	2
Room No 406	8	4
Room No 407	4	2
Room No 408	6	2
Room No 409	5	1
Room No 410	16	11
Room No 411	9	5
Room No 412	7	3
4 th Floor & Toilet	13	9
Fifth Floor		
Room No 501	6	2
Room No 502	10	5
Room No 503	12	8
Room No 504	4	2
Room No 505	4	2
Room No 506	6	3
Room No 507 A/B	6	2
Room No 508	4	2
Room No 509	4	2
5 th Floor & Toilet	15	10
Sixth Floor		
Room No 601	6	2
Room No 602	15	9
Room No 603	19	13
Room No 604	4	1
Room No 605	6	2
Room No 606	15	9
Room No 607 A & B	7	3
Room No 608	15	9
Room No 609	4	2
6 th Floor & Toilet	15	10
Seventh Floor		
Room No 701	6	2
Room No 702	16	10
Room No 703	8	4
7 th Floor & Toilet	7	3
Eighth Floor		
Room No 8th Floor	34	26
8 th Floor & Toilet	7	3

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Ceiling Fans:

Location	Quantity (No.)
Ground Floor	
Room No 001	9
Room No 002	9
Room No 004,005	2
Room No 006	11
Room No 009	9
Ground Floor Passage & Bathroom	4
First Floor	
Room No 101	9
Room No 102	13
Room No 103	9
Room No 106	9
Room No 107	11
Room No 108	9
Second Floor	
Room No 201	9
Room No 202	12
Room No 203	9
Room No 204	31
Room No 205	6
Room No 206	12
Room No 207	9
Third Floor	
Room No 301	9
Room No 302	12
Room No 303	6
Room No 304	4
Room No 305	6
Room No 306	4
Room No 307	7
Room No 308	4
Room No 309	7
Room No 310	9
Room No 311	12
Room No 312	9
3 rd Floor	2
Fourth Floor	
Room No 401	7
Room No 402	12
Room No 403	9
Room No 405	7
Room No 407	7
Room No 409	7
Room No 410	8
Room No 411	12
Room No 412	6
4 th Floor	3
Fifth Floor	
Room No 501	8

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Location	Quantity (No.)
Room No 502	12
Room No 503	8
Room No 504	4
Room No 505	9
Room No 506	6
Room No 507 A/B	6
Room No 508	4
Room No 509	9
Sixth Floor	
Room No 601	8
Room No 602	12
Room No 603	9
Room No 604	9
Room No 605	9
Room No 606	6
Room No 607 A & B	10
Room No 608	5
Room No 609	7
Seventh Floor	
Room No 701	9
Room No 702	8
Room No 703	9
7 th Floor & Toilet	EX 2
Eighth Floor	
Room No 8th Floor	8
8 th Floor	2

Air Conditioning Units:

Location	Type	Capacity (TR)	Quantity (Nos.)
Apple Room	Split	3	2
Staff Room	Split	3	1
Conference Room	Split	3	1
Startup Room	Split	2	1
Reception Passage	Split	1	2

Observations & Suggestions:

- A few of the light fittings are provided with high efficiency LED lamps. The conventional tube lights operating for longer period should be replaced with energy efficient LED lamps.
- The Air Conditioning Units are with 3 star rating, which are intrinsically operate at high energy efficiency.
- The Air Conditioners are operated as required with manual control. The operation is minimal consequently automation may not be economical.
- The temperature is maintained at 24 to 25 °C for air conditioning spaces, which is within the recommended values.
- The fans are of standard efficiency; which may be replaced with high efficiency fans with BLDC motor.
- The rooms are well ventilated and provided with fans at appropriate location for proper air circulation.

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- The gadgets are services properly and maintained in good condition.

Electricity Bills:

Consumer Name: Sardar Patel Institute of Technology					Service Provider: Adani Electricity		
Tariff Category: LT IX (B)					Consumer No. 102694354		
Meter No.: L985263					Contract Demand: 180 KVA		
Description	Unit	Apr-18	May-18	Jun-18	Jul-18	Aug-18	Sep-18
Energy Consumption	KWH	28390	20180	12982	19562	27624	21224
Bill	Rs	259480	179420	139506	201260	300564	244070
Cost	Rs/KWH	9.14	8.89	10.75	10.29	10.88	11.50
Description	Unit	Oct-18	Nov-18	Dec-18	Jan-19	Feb-19	Average
Energy Consumption	KWH	34996	17207	10154	17694	19119	20830
Bill	Rs	384990	201350	113821	202870	221370	222609
Cost	Rs/KWH	11.00	11.70	11.21	11.47	11.58	10.69

Consumer Name: Sardar Patel Institute of Technology					Service Provider: Adani Electricity		
Tariff Category: LT IX (B)					Consumer No. 152372652		
Meter No.: L1001405					Contract Demand: 129.60 KVA		
Description	Unit	Apr-18	May-18	Jun-18	Jul-18	Aug-18	Sep-18
Energy Consumption	KWH	5477	5964	5608	5516	4526	5375
Bill	Rs	61716.81	66210	62810	62380	50280	62750
Cost	Rs/KWH	11.27	11.10	11.20	11.31	11.11	11.67
Description	Unit	Oct-18	Nov-18	Dec-18	Jan-19	Feb-19	Average
Energy Consumption	KWH	6970	5518	4567	3865	3545	5176
Bill	Rs	78160	65430	52521	44950	41510	58974
Cost	Rs/KWH	11.21	11.86	11.50	11.63	11.71	11.39

Consumer Name: Sardar Patel Institute of Technology					Service Provider: Adani Electricity		
Tariff Category: LT XI (B)					Consumer No. 152372205		
Meter No.: L1001160					Contract Demand: 127.80 KVA		
Description	Unit	Apr-18	May-18	Jun-18	Jul-18	Sep-18	
Energy Consumption	KWH	5750	3640	1490	3189	5473	
Bill	Rs	64235.33	40160	16310	36190	64610	
Cost	Rs/KWH	11.17	11.03	10.95	11.35	11.81	
Description	Unit	Oct-18	Nov-18	Dec-18	Feb-19	Average	
Energy Consumption	KWH	6988	2960	1064	3416	3774	
Bill	Rs	77900	35860	12470	40600	43148	
Cost	Rs/KWH	11.15	12.11	11.72	11.89	11.43	

Observations & Suggestions:

- The power consumption & cost is within the normal range.

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IV Environmental System

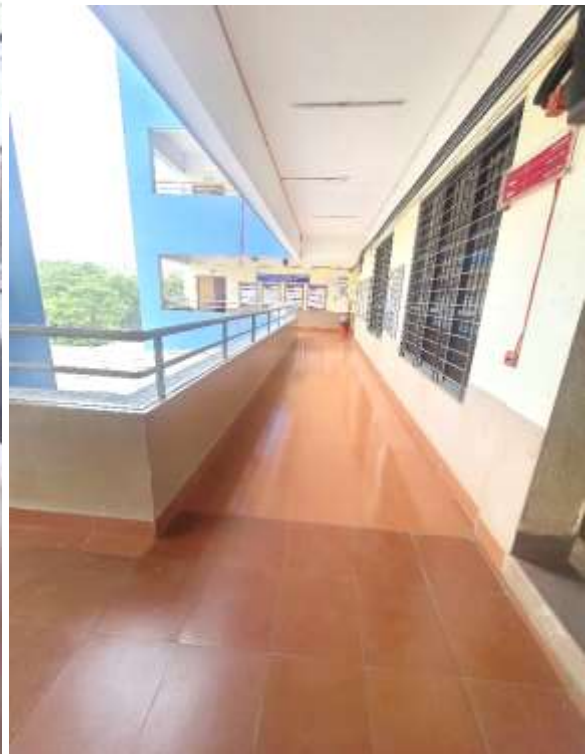
Ventilation & Air Quality:

- The air ventilation is adequate.
- Several indoor & outdoor plants have been installed to improve air quality.



- The air conditioned rooms are provided with proper ventilation and fresh air.
- It has been a general practice to switch off the fans & lights in an unoccupied area.

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V Water Management

Water Source:

The water supplied by the municipal corporation is used for drinking and other requirements. The incoming water from the municipal corporation is metered.



The water consumption details are as under.

Period	Days	Consumption	Bill Amount	Cost
		KL	Rs	Rs/KL
Connection No- KW@0359909				
16-04-2018 to 17-05-2018	31	521	4427	8.50
17-05-2018 to 18-06-2018	32	538	4926	9.16
18-06-2018 to 17-07-2018	29	488	4223	8.65
17-07-2018 to 18-08-2018	32	538	4829	8.98
19-10-2018 to 19-11-2018	31	522	4787	9.17
19-11-2018 to 19-12-2018	30	505	4548	9.01
19-12-2018 to 19-01-2019	31	522	4702	9.01
19-01-2019 to 20-02-2019	32	539	4850	9.00

Observation & Suggestions:

- The cost as well as consumption is within the normal range.

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Water Coolers & Purifiers:

Sr No	Location	Quantity (No.)
1	Ground Floor	2
2	First Floor	1
3	Second Floor	1
4	Third Floor	1
5	Fourth Floor	1
6	Fifth Floor	1
7	Sixth Floor	1

The water coolers & purifiers are provided at convenient locations.

Water Distribution System:

The distribution network and piping are more or less satisfactory and adequate.

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Rainwater Harvesting:

The rainwater is fed to the rainwater harvesting pit for ground water recharge.



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VI Waste Generation & Management

Solid Waste:

- The organic as well as inorganic waste is segregated in the college premises.
- The organic waste is used to generate manure by composting.



- There is no generation of hazardous or biomedical waste in the premises.
- Each Floor is provided is with segregated dustbin for Recyclable waste and Non-Recyclable Waste



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Sewage & Wastewater:

- The sewage is disposed through municipal system.
- There is no generation of harmful or hazardous effluent.
- The sewage treatment plant is not required due to minimal generation of waste.
- The municipal corporation charges cess for treating and disposing wastewater.

VII Infrastructure & Safety



Observations:

- The premises are provided with multiple staircases with necessary entrances to ensure quick and effective movement in normal as well as emergency conditions.



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- The movement of vehicle inside the campus is restricted with vehicles of Special Dignitaries and VIP Guests are allowed to enter the campus.
- There are pedestrian friendly pathways for in-campus movement.
- There is complete ban plastic usage inside the campus.
- There is adequate parking space for vehicles/four wheelers. However, almost all the students and many of the faculty members avail public transport system which is very convenient due to proximity to railway station and bus services.
- Ramp & wheelchair is available for differently abled student.

Draining system:

- The drains from the washrooms are connected to the municipal drainage, which is a common practice in the colleges in and around Mumbai.

Seepage in the building:

- The premise was visually inspected for seepages. No seepages were observed in any of the places.

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Firefighting & fire escape system:



Sr No	Location	Type	Capacity	Quantity (No.)
1	Ground Floor	ABC Dry Powder	6 KG	5
2	First Floor	ABC Dry Powder	6 KG	2
		ABC Safe Zone	5 KG	3
3	Second Floor	ABC Dry Powder	6 KG	7
		ABC Safe Zone	5 KG	1
		Water	9 LTR	2
4	Third Floor	ABC Dry Powder	6 KG	5
		Water	9 LTR	1
5	Fourth Floor	ABC Dry Powder	6 KG	5
		ABC Safe Zone	5 KG	1
6	Fifth Floor	ABC Dry Powder	6 KG	1
		ABC Safe Zone	5 KG	3
		Water	9 LTR	1
7	Sixth Floor	ABC Dry Powder	6 KG	4
		Water	9 LTR	1
8	Seventh Floor	ABC Dry Powder	6 KG	1
9	Eighth Floor	ABC Dry Powder	6 KG	2

- There are efficient fire extinguishers in the premises, which are checked / refilled as per the stipulated frequency.
- The premise is provided with multiple staircases with requisite entrances to ensure quick and effective movement in emergency conditions.

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Parking space:



- There is adequate parking space for vehicles/four wheelers. However, almost all the students and many of the faculty members avail public transport system which is very convenient due to proximity to railway station and bus services.

VIII Green Culture



Observations & Suggestions:

- The LED / LCD monitors & Laptops has been procured, which are energy efficient.
- These monitors are not only energy efficient but also generate minimal heat and cut down on air conditioning load.
- The electronic communication may be encouraged to minimize usage of papers.
- Most of the papers are reused for doubled sided printing to further minimize usage of paper.

The following steps may be initiated to further enhance efficiency of the systems.

1. An efficient power management system may be incorporated to
 - a. Switch off the display if not in use.
 - b. Put the computer in Sleep mode / switching off the machines, if not used for prolonged period.
2. Optimize brightness of the screen.
3. Discourage use of screen savers, which has similar power consumption.

Re-using one sided paper for printing:

It was observed that two side printing / printing on the back side of used paper in more than 80% of the cases.

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IX Renewable Energy

Solar Photovoltaic:

It is suggested to install rooftop solar panels with net metering for captive usage.

Solar Thermal:

There is no application of solar thermal system and does not find attractive in this case.

Biogas Plant:

There is no possibility of installing biogas plant for cooking as the quantity of plate waste is negligible.